Musanze Harvest Intelligence Service

AI decision-support prototype for a potato cooperative - Applied AI individual project

Item	Value
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Verification code	**AI-A2-7889**
Dataset seed	7889
Repository / folder	AI_A2_2527889

Stakeholder

The stakeholder is the management of a potato cooperative in Musanze, together with its extension staff. They need a quick, offline-capable way to plan collection, storage and field visits before harvest.

Decision problem and prediction questions

1. **Regression:** how many tons of potatoes is a farm likely to harvest (yield_tons)?
2. **Classification:** is a farm at high production risk, meaning its yield per hectare is below the median (high_risk = 1)?
3. **Clustering:** which farms share similar growing conditions and could be advised or supported together?

Ethical boundary

This is a classroom prototype trained on **synthetic data**. It does not replace an agronomist and must not be used to decide access to finance, insurance or public services. Every output needs human review. Training and the command-line predictor run locally, so they work when internet access is unreliable; the FastAPI service exposes the same predictions for later integration with a mobile application.

Data and reproducibility

Data dictionary

Column	Rule / type	Role
rainfall_mm	Uniform 600-1400	Numeric feature
avg_temp_c	Uniform 12-26	Numeric feature
soil_moisture_pct	Uniform 20-80	Numeric feature
fertilizer_kg	Uniform 0-250	Numeric feature
field_size_ha	Uniform 0.2-4.0	Numeric feature
season	Random A or B	Categorical feature
cooperative_support	Random 0 or 1	Binary feature
yield_tons	Formula + N(0, 0.80) noise, clipped at 0.50	Regression target
high_risk	Yield per hectare below its median	Classification target

Personal seed rule

Seed = 7889 (registration digits), used as `numpy.random.default_rng(7889)`. Base rows = 240 + (7889 mod 41) = 240 + 17 = **257**. Missing values: m = round(0.05 x 257) = 13 in each of soil_moisture_pct and fertilizer_kg (different positions), then 3 exact duplicate rows are appended, giving **260 raw rows**. The CSV uses four decimals (float_format="%.4f"). The yield formula is: 0.004 x rainfall + 0.06 x soil moisture + 0.02 x fertilizer + 1.30 x field size - 0.25 x |temperature - 18| + 0.70 x cooperative support + season effect (+0.50 A, -0.20 B) + noise.

Data-quality evidence (artifacts/validation_report.json)

Check	Result
Raw rows	260
Missing soil_moisture_pct	13
Missing fertilizer_kg	13
Exact duplicate rows detected and dropped	3
Rows after cleaning	257
Range / label checks	passed
Target leakage check	passed (no yield_tons, yield per hectare or high_risk as input)

SHA-256 fingerprint of data/AI_A2_2527889.csv

`dedec532494f23dd25fe7747ed497996b86c368459eff63c75494dc7c915f1aa`

Running run_all.py twice recreates the same CSV bytes and therefore the same fingerprint; the automated reproducibility test checks this.

Pipeline design

Train/test separation

Duplicates are removed first so identical rows cannot appear on both sides. The 257 cleaned rows are split once with train_test_split(test_size=0.25, random_state=7889, stratified on high_risk): **192 training rows and 65 test rows**. The test set is used only for final reporting; model selection uses 5-fold cross-validation inside the training set.

Preprocessing flow (one ColumnTransformer, reused at inference)

Inputs	Steps (fitted on training data only)
rainfall_mm, avg_temp_c, soil_moisture_pct, fertilizer_kg, field_size_ha	median imputation, then StandardScaler
season	most-frequent imputation, then one-hot encoding
cooperative_support	most-frequent imputation (already 0/1)

The preprocessor and the estimator are combined in a scikit-learn Pipeline and saved with joblib, so training, cross-validation and inference apply identical transformations. Because the imputer sits inside the Pipeline, cross-validation re-fits it on each fold's training part.

Leakage controls

Model inputs are exactly: rainfall_mm, avg_temp_c, soil_moisture_pct, fertilizer_kg, field_size_ha, season, cooperative_support. yield_tons, yield per hectare and high_risk are never inputs for the classifier; src/validation.py raises an error if such a column is added. No imputer, encoder, scaler or feature selection is fitted before the split. High_risk is defined from yield per hectare, which is why yield_tons is excluded from the classifier's inputs; field_size_ha is a legitimate input known before harvest.

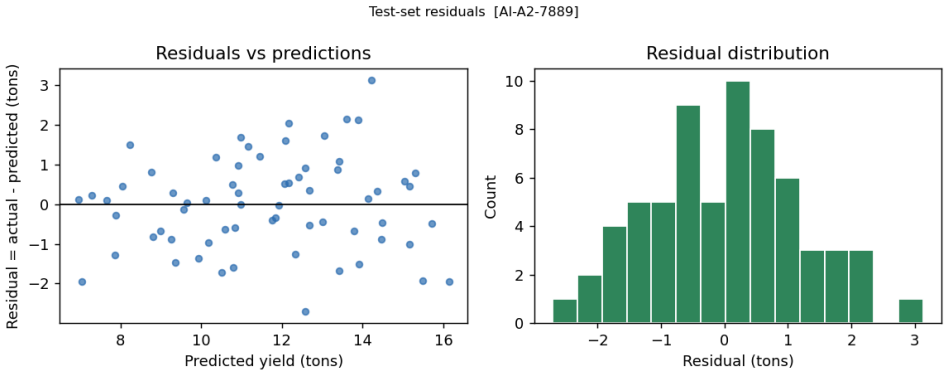
Model candidates

Task	Baseline	Candidates	Selection rule
Regression	Mean predictor	Linear regression, random forest, gradient boosting	lowest 5-fold cross-validated RMSE on the training set
Classification	Majority class	Logistic regression, random forest, gradient boosting	highest 5-fold cross-validated ROC-AUC on the training set
Clustering	-	K-means, K = 2..6 on scaled inputs	highest silhouette score among K = 2..6

All random states are set to 7889 and recorded in the JSON artifacts. The final models are fitted on the training rows only, so the reported test metrics describe exactly the saved models.

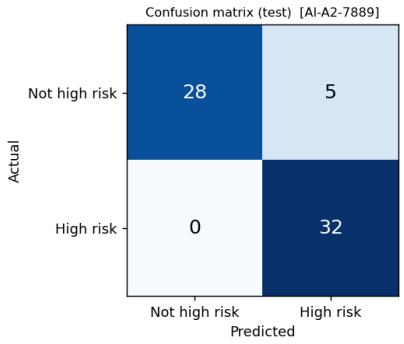
Regression and classification

Model (regression)	MAE (t)	RMSE (t)	R^2	CV RMSE
mean_baseline	2.3013	2.7634	-0.0222	-
linear_regression (selected)	0.9468	1.1752	0.8151	1.0018
random_forest	1.1928	1.4895	0.7030	1.2510
gradient_boosting	1.1200	1.4380	0.7232	1.1628



Selected regression model. linear_regression had the lowest cross-validated RMSE (1.0018 t). On the held-out test set it scored MAE 0.9468, RMSE 1.1752, R2 0.8151 versus the mean baseline MAE 2.3013, RMSE 2.7634, R2 -0.0222. Test residuals (actual - predicted): mean +0.0058 t, standard deviation 1.1843 t, largest absolute error 3.1341 t.

Model (classification)	Precision	Recall	F1	ROC-AUC	CV AUC
majority_baseline	0.0000	0.0000	0.0000	0.5000	-
logistic_regression (selected)	0.8649	1.0000	0.9275	0.9811	0.9853
random_forest	0.8438	0.8438	0.8438	0.9631	0.9696
gradient_boosting	0.8710	0.8438	0.8571	0.9650	0.9556



Selected classifier. logistic_regression had the highest cross-validated ROC-AUC (0.9853). On the held-out test set it scored precision 0.8649, recall 1.0000, F1 0.9275, ROC-AUC 0.9811; the majority baseline scored F1 0.0000 and ROC-AUC 0.5000.

Confusion matrix (rows actual, columns predicted): TN 28, FP 5, FN 0, TP 32. A false negative (a high-risk farm missed) is the costlier error for a cooperative planning support, so recall matters most here.

Important errors (my interpretation)

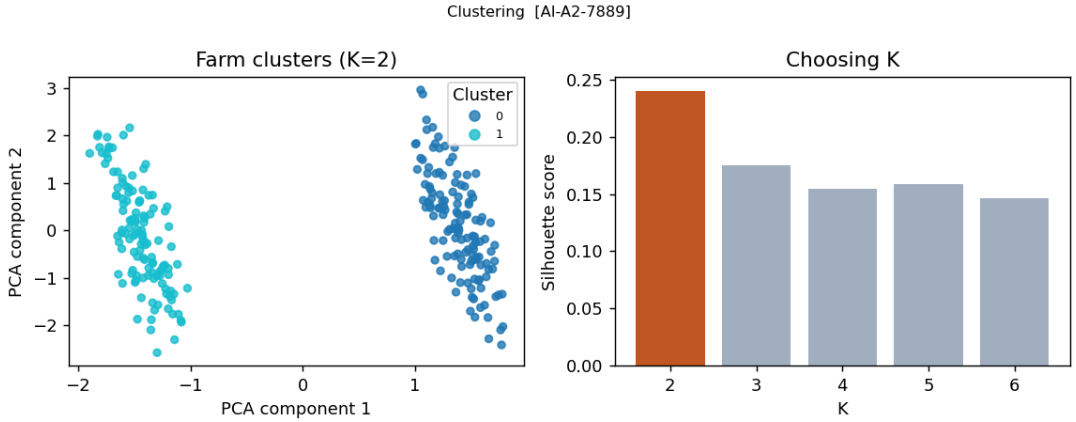
Regression: Row 208 had the largest test error: actual 17.3608 t, predicted 14.2267 t (error -3.1341 t, 3.9 times the generator's noise standard deviation of 0.80 t). Inputs: rainfall_mm=1308.17, avg_temp_c=18.00, soil_moisture_pct=51.31, fertilizer_kg=71.77, field_size_ha=3.23, season=A, cooperative_support=1. The generator adds random noise the model cannot observe. The selected linear model fits temperature as a straight line, while the generator penalises distance from 18 C (an absolute value); this row's temperature is 18.00 C, so this mismatch may have contributed (not tested).

Classification: Row 54: actual class 0, predicted 1 with probability 0.8349. Its yield per hectare is 6.066 t/ha against a dataset median of 5.640 t/ha, so it lies 0.427 t/ha from the class boundary. Total misclassifications on the test set: 5 of 65.

Clustering

Inputs are imputed, encoded and scaled (fitted on training rows), then K-means is run for K = 2 to 6 (n_init = 10, random_state = 7889). Silhouette score decides K.

K	Silhouette	Inertia	Chosen
2	0.2403	1147.5	yes
3	0.1754	1056.0	
4	0.1541	960.5	
5	0.1585	906.6	
6	0.1464	849.0	



Selected K = 2. K=2 gave the highest silhouette score (0.2403). Cluster structure is weak: the groups overlap and should be read as soft segments. Yield and risk rates are shown for description only; they were not used to form clusters. The data are synthetic and drawn from uniform distributions, so clusters are convenient groupings of similar conditions, not natural farm types.

Cluster	Farms (train)	Mean yield (t)	High-risk rate	Plain-language profile
0	101 (53%)	12.54	46%	mostly season A
1	91 (47%)	11.48	55%	mostly season B

Operational meaning

Cluster labels are arbitrary numbers fixed by the seed. Cluster 0 is the largest (101 training farms, 53%): mostly season A. Mean yield 12.54 t, high-risk rate 46%, mean rainfall 1031 mm, mean fertilizer 126 kg, mean field size 2.06 ha. Operationally, such a group could be visited or advised together, with human review. Yield and risk columns describe each group afterwards; they were not used to form it.

API and testing

Endpoint	Purpose	Responses
GET /health	liveness and whether models are loaded	200
GET /model-info	selected models, test metrics, fingerprint, verification code	200; 503 if models missing
POST /predict	yield estimate, risk class and probability, cluster	200; 422 invalid input; 503 if models missing

Valid request and response

```
rainfall_mm: 1050.0, avg_temp_c: 18.5, soil_moisture_pct: 55.0
fertilizer_kg: 140.0, field_size_ha: 1.8, season: "A"
cooperative_support: 1
=> HTTP 200
verification_code: "AI-A2-7889", yield_estimate_tons: 12.8905, high_risk_prediction: 0
high_risk_label: "not high risk", high_risk_probability: 0.0986, cluster: 0
cluster_profile: "mostly season A"
```

Invalid request (out-of-range, wrong type, missing field, bad category) returns HTTP 422

```
rainfall_mm: 2500.0, avg_temp_c: "warm", soil_moisture_pct: 55.0
fertilizer_kg: 140.0, season: "C", cooperative_support: 1
=> HTTP 422
rainfall_mm: Input should be less than or equal to 1400
avg_temp_c: Input should be a valid number
field_size_ha: Field required
season: Input should be 'A' or 'B'
```

Failure behaviour

Inputs are strictly typed (pydantic): all seven fields are required, numbers must lie in the training ranges, season must be A or B, cooperative_support must be 0 or 1, and unknown fields are rejected. Errors list each failing field and never crash the server. The CLI prints the same details to stderr and exits with code 2. If models are missing, the API returns 503 with instructions to run run_all.py.

Automated tests (pytest)

Test	Covers
test_valid_inference	valid record returns all outputs with the verification code
test_missing_field_rejected	missing field raises a validation error
test_wrong_type_rejected	string given for a numeric field is rejected
test_range_error_rejected	out-of-range value is rejected
test_reproducibility	two full pipeline runs give the same fingerprint and metrics
test_api_predict_response	POST /predict returns 200 and the expected keys
test_api_invalid_returns_422	API returns a useful 422 for invalid input
test_api_health_and_model_info	/health and /model-info respond with the code
test_leakage_guard	target columns are refused as model inputs
test_run_all_rejects_wrong_seed	run_all.py refuses any seed other than 7889

Last recorded run: 10 passed, 1 warning in 16.92s.

Model card and reflection

Section	Statement
Intended use	Classroom decision-support prototype: rough harvest estimates, flagging farms for follow-up and grouping similar farms, always with human review.
Out of scope / prohibited	Replacing an agronomist; deciding loans, insurance, subsidies or access to public services; any use on real farmers' data without new validation.
Data	Fully synthetic (260 raw / 257 cleaned rows), generated from a known formula. No real farm was measured.
Performance	Test MAE 0.95 t, RMSE 1.18 t, R^2 0.82; risk F1 0.93, ROC-AUC 0.98, recall 1.00, precision 0.86 on 65 test farms.
Limitations	High scores partly reflect that the data come from a simple linear rule plus Gaussian noise; real farms involve pests, disease, soil type, variety and weather timing. The test set is small (65 rows), so metrics have wide uncertainty. Inputs outside the training ranges are rejected, not extrapolated. Clusters are weak (low silhouette).
Risks	Over-trust in a number; applying results to real farms; unfair treatment if risk labels were reused for credit or support decisions.
Human oversight	A cooperative officer or agronomist reviews each output and can override it. Predictions are advice, not decisions.
Improvements with real data	Collect measured yields and field records, validate on a separate season or district, add uncertainty intervals, check performance by group, and retrain with agronomist input.

Reflection

The linear model beat the tree ensembles on cross-validation, which fits the way the data were generated; on real data I would not expect that. Running every stage from one command made mistakes such as duplicate rows and imputation order visible early. Evidence for each claim sits in the JSON, CSV and plot artifacts listed in the README.